

Homework — 2.1.3 Thinking Procedurally

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 2.1.3 lesson.

Part A — This week's topic (~14 marks)

1. A team is building an online cinema-ticket booking system. Identify **three** sub-procedures that the problem could be decomposed into. **[3]** (AO2)

2. In the booking system, three of the steps are: `selectSeats` → `takePayment` → `issueTicket`. Explain why `issueTicket` must come **after** `takePayment`, referring to the output of the earlier step. **[2]** (AO2)

3. A washing-machine control program runs these steps: lock door → fill with water → heat water → wash → drain → spin.

(a) Identify **one** pair of these steps whose order is fixed, and explain the dependency between them. **[2]** (AO2)

(b) The program also updates a countdown display. Explain why the order of this task relative to the wash step is **not** fixed. **[1]** (AO2)

4. A program for an online quiz calls the same `gradeAnswer` sub-procedure for every question. Explain **two** advantages of writing this once as a sub-procedure rather than copying the grading code under each question. [2] (AO2)

5. A games company has four programmers building one large open-world game. Explain how **decomposition** allows the programmers to work effectively as a team. [2] (AO2)

6. A council needs software to process residents' parking-permit applications.

Identify a suitable sequence of **named** sub-procedures for this problem, and justify why the order of **one** pair of your steps is fixed, referring to the output of the earlier step. [2] (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

7. (2.1.2) State what is meant by a **precondition**, and give one example of a precondition for a routine that performs a **binary search**. [2] (AO1/AO2)

8. (1.2.2) State, in order, **two** stages of compilation that come **before** code generation. [2] (AO1)

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9. (1.2.3) State **one** advantage of the **Agile** development methodology over the **Waterfall** lifecycle model.
[2] (AO1)

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